

Lucas Root

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Computer Science student (Cybersecurity & AI threads) at the Georgia Institute of Technology, seeking a software engineering internship.

EDUCATION

Georgia Institute of Technology

B.S. in Computer Science: Cybersecurity & AI

Atlanta, GA

Transferred in Aug. 2026 - May 2028

University of Kansas

B.S. in Computer Science & Cybersecurity Engineering, Honors College

Lawrence, KS

Aug. 2024 - Transferred out May 2026

– 3.9/4.0 GPA — Coursework: Programming 2, Software Engineering, Discrete Structures, Embedded Systems

EXPERIENCE

Air Force Research Laboratory (AFRL/RI)

AI/ML Research Intern | ZEIOS Vanguard Group | VICEROY MAVEN

Rome, NY

Jun. 2026 - Aug. 2026

- Contributed to 5 privacy and AI research initiatives over an 8-week internship.
- Engineered privacy-preserving metadata obfuscation pipelines for **SWEEP & SHADOW** using Python, OHTTP/ODoH protocols, and mitmproxy, reducing request traceability across test suites.
- Evaluated LLM-driven Private Information Retrieval architectures for **WHISPER**, benchmarking local LLM query latency and throughput against live weather APIs.
- Calculated inter-annotator agreement using Krippendorff's Alpha for **SCOPE** video annotation tasks to establish baseline dataset quality for downstream vision models.
- Analyzed real-time streaming video-language model performance under continuous input constraints for **RANGER**.

PROJECTS

TURT: Custom Unix Shell | C, Linux System Calls, Makefile

Oct. 2025

- Engineered a POSIX-compliant shell in C, managing process execution via `fork`, `execvp`, and `wait`.
- Built a custom tokenizer and signal handling for built-in commands, replicating core Bash functionality.

InternSweep | Python, Playwright, SQLite, Streamlit

2026

- Built an automated internship discovery pipeline that scrapes Greenhouse and Lever boards, tracks application statuses, and presents reviewable listings in a centralized dashboard.

Joumé | Rust, SQLite, Stripe, Leptos

Dec. 2024 - Present

- Architected an end-to-end e-commerce platform for an independent clothing brand featuring secure payment integration and local database inventory tracking.

Praxis | Rust, Next.js, Postgres

Jan. 2026 - Present

- Architected a full-stack student collaboration platform with a type-safe Rust backend and relational PostgreSQL schema.
- Designed reactive user workflows and project feeds to connect student developers for technical teamwork.

Get Big, Man | C#, Unity, FL Studio

Feb. 2025

- Engineered a 2D roguelike game addressing gym anxiety, winning Best Beginner Project at HackKU 2025.
- Implemented core game mechanics, player movement loops, and procedural generation logic within a 36-hour development window.

LEADERSHIP & EXTRACURRICULARS

Rowing Experience & Leadership

Aug. 2019 - Present

NorCal Crew, Kansas Crew, Georgia Tech Crew

- Varsity Rower, Georgia Tech Crew (Aug. 2026 - Present)
- President, Kansas Crew (Aug. 2024 - May 2026): Managed team logistics, recruitment, and community partnerships for the official university rowing club.
- Varsity Rower, NorCal Crew (Aug. 2019 - May 2024): Competed nationally, placing 13th at the 2023 Head of the Charles; coached novice rowers ages 11-15.

HONORS & AWARDS

KU Excellence Scholarship

2024

University of Kansas

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C, C++, Rust, C#

Developer Tools: Git/GitHub, Docker, Linux/Unix, GDB, Make, VSCode

Concepts: Memory Management, OS Architecture, Embedded Systems, Data Structures, Algorithms